

FATMA ADEL MOHAMED

Backend Developer (Java & Spring) — Cairo, Egypt

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Objective

Fourth-year CS student specialized in backend development (Java & Spring) with experience in AI-powered applications using LLMs and RAG systems. Seeking a backend or full-stack internship opportunity.

Education

Bachelor's Degree in Computer Science – Al-Obour High Institute, Cairo

2022 – 2026

Technical Skills

- **Languages:** Java, Python, C++, SQL
- **Backend & Frameworks:** Java SE, Java EE, Spring Boot, Spring MVC, Spring Security, Microservices
- **AI & LLM:** LangChain4j, OpenAI API (GPT-4o-mini), RAG Systems, ChromaDB, Vision AI
- **APIs & Security:** RESTful APIs, JWT Authentication, Swagger, Postman
- **Database & Persistence:** MySQL, JDBC, Hibernate ORM, Spring Data JPA
- **Front-End:** HTML5, CSS3, JavaScript (Basics), JSP, Angular (Basics)
- **Testing & Tools:** JUnit5, Maven, Git, GitHub, Docker, IntelliJ IDEA, VS Code
- **Concepts:** OOP, Data Structures, Algorithms, MVC, Layered Architecture, NLP (Bilingual AI)

Projects

Animal AI Chatbot (RAG + Vision)

Spring Boot, LangChain4j, OpenAI GPT-4o-mini, ChromaDB, Docker

- Built an AI-powered chatbot using RAG with ChromaDB vector store for accurate animal information retrieval.
- Integrated OpenAI Vision API to analyze uploaded animal images (gender, age, health condition).
- Implemented bilingual NLP support (Arabic & English) with automatic language detection.
- Designed modular service layer (VisionService, AnimalChatService, TextAnimalService) via AnimalChatFacade.
- Containerized ChromaDB with Docker; exposed REST endpoints for image upload and text queries.

ToDo App Backend (Microservices)

Spring Boot, Spring Security, JWT, MySQL, Swagger, JUnit5

- Developed User Service and Todo Service with JWT-based authentication and inter-service token validation.
- Added CRUD operations, search, validation, global exception handling, and OTP-based password reset via Java Mail.
- Documented APIs using Swagger and Postman.

CRUD Management System

Spring MVC, Hibernate, MySQL, JSP

- Developed a full CRUD web application following MVC architecture with Hibernate ORM and JSP-based UI.

Activities

- Self-training in backend development and AI integration using Spring Boot and LangChain4j.
- Solved algorithmic problems on Codeforces and HackerRank.

Languages

Arabic (Native)

English (Proficient)